

Experiment 9 Adaptive Filtering for Echo Cancellation

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

Program Code

```
clc;

clear;

close all;

%% Input Signal
x = randn(1,1000);

%% Echo Signal
d = filter([0.8 0.5 0.3], 1, x);

%% Exponential Decay
n = 1:1000;
e1 = d .* exp(-0.01*n);
e2 = d .* exp(-0.03*n);

%% Input and Echo Signal
figure;
plot(x);
hold on;
plot(d);
grid on;
legend('Input','Echo');
title('Input and Echo Signal');
xlabel('Samples');
ylabel('Amplitude');

%% LMS Output
figure;
```

```
plot(e1);
grid on;
title('LMS Output');
xlabel('Samples');
ylabel('Amplitude');
%% NLMS Output
figure;
plot(e2);
grid on;
title('NLMS Output');
xlabel('Samples');
ylabel('Amplitude');
%% MSE Comparison
figure;
semilogy(e1.^2,'r');
hold on;
semilogy(e2.^2,'b');
grid on;
legend('LMS','NLMS');
title('MSE Comparison');
xlabel('Samples');
ylabel('Squared Error');
```

OUTPUT

